# Guangyu Xiang

Ph.D. Student in Data Science and Analytics · HKUST(GZ)

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Efficient and elastic LLM / multimodal training and inference · GPU computing

## Education

**The Hong Kong University of Science and Technology (Guangzhou)** Feb. 2025 - Present

Ph.D. in Data Science and Analytics, Information Hub · Guangzhou, China

Advisor: Prof. Xiaowen Chu. Research interests: machine learning systems, distributed training and inference, GPU computing, and communication-computation co-optimization.

**Peking University** Sep. 2021 - Jun. 2024

Master's Degree in Software Engineering · Beijing, China

School of Software and Microelectronics. Research focus: high-performance computing, AI compiler optimization, and CUDA programming.

**University of Electronic Science and Technology of China** Sep. 2017 - Jun. 2021

Bachelor's Degree in Software Engineering · Chengdu, China

School of Information and Software Engineering. GPA: 3.86/4.00; ranked in the top 5%. National Encouragement Scholarship; Outstanding Student Scholarship.

## Publications

* Equal contribution.

**ElasWave: An Elastic-Native System for Scalable Hybrid-Parallel Training**

Xueze Kang*, **Guangyu Xiang***, Yuxin Wang, Hao Zhang, Yuchu Fang, Yuhang Zhou, Zhenheng Tang, Youhui Lv, Eliran Maman, Mark Wasserman, Alon Zameret, Zhipeng Bian, Shushu Chen, Zhiyou Yu, Jin Wang, Xiaoyu Wu, Yang Zheng, Chen Tian, and Xiaowen Chu

IEEE HPCA 2027 · Accepted

**Zellige: Moldable Sequence Placement for Mixed Image-Video DiT Training**

**Guangyu Xiang***, Xueze Kang*, Minwei Zhao, Yuxin Wang, Shaohuai Shi, Lin Zhang, and Xiaowen Chu

arXiv preprint · 2026

**Xema: Efficient Diffusion Serving through Fine-Grained Memory Management and Auto-Configuration**

Xueze Kang*, **Guangyu Xiang***, Suyi Li, Yuxin Wang, Shaohuai Shi, Lin Zhang, and Xiaowen Chu

arXiv preprint · 2026

**KernelFlume: Elastic Core-Attention Scaling for Agentic Long-Context Decoding**

**Guangyu Xiang***, Xueze Kang*, Lin Zhang, Wenxiang Lin, Shaohuai Shi, Yuxin Wang, and Xiaowen Chu

arXiv preprint · 2026

**Compass: Dissecting Communication and Computation Operators for Efficient LLM Training**

**Guangyu Xiang**, Lin Zhang, Haoxuan Yu, Xinglin Pan, Shaohuai Shi, and Xiaowen Chu

IEEE INFOCOM 2026

## Research and Engineering Experience

**Microsoft Research Asia (MSRA)** Feb. 2023 - Dec. 2023

Research Intern, Systems Research Group · Beijing, China

Analyzed FlashAttention computation, memory access, GPU execution, and CUDA/CUTLASS implementations. Developed CUDA kernels and explored loop transformations for data locality and compiler cost models.

**OneFlow** Sep. 2022 - Feb. 2023

R&D Engineer Intern, AI Framework Team · Beijing, China

Migrated MLIR operator-fusion patterns from TableGen DRR to PDLL. Developed cross-framework model migration and validation workflows for ResNet-50 and GPT-2, including execution and result-consistency checks on Cambricon MLU.

**AMD** Mar. 2022 - Sep. 2022

AI Compiler Engineer Intern · Beijing, China

Developed TVM Relay passes for ASR deployment on AMD AI Engine, replacing quantize/dequantize chains with native quantized operators. Participated in hardware testing, deployment debugging, accuracy validation, and performance evaluation.

**Peng Cheng Laboratory** Nov. 2020 - Aug. 2021

AI Development Engineer Intern · Shenzhen, China

Contributed to resource management and job scheduling for high-performance computing clusters.

## Projects

### Communication-Computation Fusion for Collective Operations

Built topology-aware performance models from GPU peer-to-peer bandwidth measurements. Designed fused All-Gather+GEMM and GEMM+Reduce-Scatter operators and pipelining mechanisms; contributed pybind11 bindings and Megatron integration for testing, autotuning, and end-to-end performance evaluation.

### AI Operator Standards and Automated Testing

Surveyed operator interfaces in mainstream AI frameworks and contributed to AI operator standards. Implemented reference operators in AITISA and developed an automated test framework.

## Technical Skills

**Programming** Python, C/C++

**GPU & compilers** CUDA/CUTLASS, MLIR/PDLL, TVM Relay

**ML systems & tooling** Megatron-LM, Docker

## Honors

- Meritorious Winner, Mathematical Contest in Modeling (MCM)
- First Prize, National College Student Information Security Competition
- First Prize, Southwest Region, China Collegiate Computing Contest - Network Technology Challenge